

Amira Mohamed Abdo

Cairo, Egypt | +20 109 692 6733 | amira16011@gmail.com | [linkedin.com/in/amira-mhmd](#) | [github.com/amira-mhmd-ml](#)

PROFESSIONAL SUMMARY

LLM Engineer specialized in designing and deploying production-grade large language model systems, including retrieval-augmented generation (RAG) pipelines, agentic AI systems, and fine-tuning workflows. Proficient in prompt engineering, system design for LLMs, and infrastructure optimization. Experienced in building scalable, cost-efficient solutions with strong emphasis on safety, observability, and responsible AI practices.

CORE EXPERTISE

LLM Fundamentals & APIs

- LLM APIs (OpenAI, Claude, Gemini) • Prompt Engineering & Design • Token Management & Rate Limiting
- Structured Outputs & Function Calling • Vision & Multimodal Inputs • Chat Completions Format

RAG & Knowledge Systems

- Vector Databases (Qdrant, Pinecone, Chroma) • Hybrid Search & Reranking • Semantic & Metadata Filtering
- Context Window Optimization • GraphRAG & Multi-Vector Retrieval • Chunking Strategies

TECHNICAL SKILLS

Languages & Frameworks

- Python (Production-level) • LangChain, LlamaIndex, LangGraph • FastAPI & REST API Design
- Pydantic & Data Validation • TypeScript/JavaScript • Node.js & Express.js

MLOps, Cloud & Infrastructure

- Experiment Tracking (MLflow, W&B) • Model Deployment & Versioning • Docker & Kubernetes
- AWS/GCP Cloud Services • CI/CD for ML (GitHub Actions) • Serverless Architecture

ADVANCED SPECIALIZATIONS

Agents & Agentic Systems

- Tool Use & Function Calling • Agent Loops & Memory Management • Planning Patterns (ReAct, Plan-Execute)
- Multi-Agent Workflows • LangGraph & Hierarchical Agents • Agent Guardrails & Self-Reflection

Fine-Tuning & Training

- LoRA & QLoRA Optimization • Dataset Preparation & Evaluation • RLHF & DPO Concepts
- Synthetic Data Generation • Model Quantization (GGUF, AWQ) • Hugging Face Transformers

PRODUCTION & SAFETY

Inference & Deployment

- FastAPI & Flask Servers • vLLM & Ollama (Local Inference) • Batching & GPU Memory Management
- Speculative Decoding & Optimization • Latency & Cost Monitoring • Model Serving Patterns

Safety & Responsible AI

- Guardrails Design & Implementation • Output Validation & PII Detection • Hallucination Mitigation
- Bias Awareness & Fairness • A/B Testing & Evaluation Frameworks • Red-Teaming Practices

SYSTEM DESIGN & ENGINEERING

Architecture Patterns

- Semantic Caching Strategies • Async Pipelines & Queue-Based Architectures • Multi-Model Routing (LiteLLM)
- Load Balancing & Failover • Graceful Degradation • Idempotency & Error Handling

Code Quality & Engineering Mindset

- SOLID Principles & Clean Code • Type Safety (mypy, Pyright) • Unit & Integration Testing
- Pre-commit Hooks & Linting • Code Review Skills • Problem Definition & Edge Case Analysis

EVALUATION & OBSERVABILITY

LLM Evaluation

- LLM-as-Judge Pattern • RAGAS Framework • Faithfulness & Relevance Metrics
- Human Evaluation Pipelines • Regression Testing for Prompts • BLEU/ROUGE Metrics

Monitoring & Logging

- LangSmith & LLM Tracing • Latency & Cost Monitoring • Feedback Loops & Iteration
- Shadow Deployment • Performance Analytics • A/B Testing for LLMs

PROFESSIONAL DEVELOPMENT

Continuous Learning & Growth

- Research Paper Analysis (ArXiv) • Open Source Contributions • Side Projects & Experimentation
- Technical Writing & Documentation • Conference Participation (NeurIPS, ICML, ICLR)
- Engineering Mindset: Problem Definition, Interface Design, Observability Planning, Cost vs Quality Tradeoffs